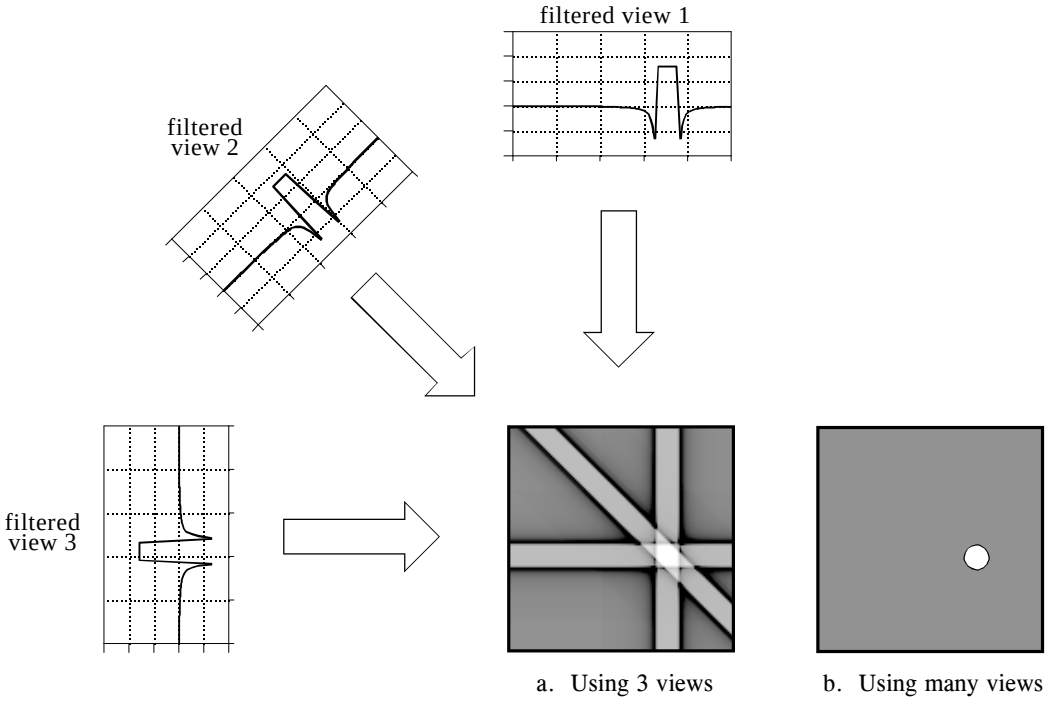

FIGURE 25-17
Filtered backprojection. Filtered backprojection reconstructs an image by filtering each view before backprojection. This removes the blurring seen in simple backprojection, and results in a mathematically exact reconstruction of the image. Filtered backprojection is the most commonly used algorithm for computed tomography systems.

to the "correct" image when there are an *infinite* number of views and an *infinite* number of points per view.

The filter kernel used in this technique will be discussed shortly. For now, notice how the profiles have been changed by the filter. The image in this example is a uniform white circle surrounded by a black background (a pillbox). Each of the acquired views has a flat background with a rounded region representing the white circle. Filtering changes the views in two significant ways. First, the top of the pulse is made flat, resulting in the final backprojection creating a *uniform* signal level within the circle. Second, negative spikes have been introduced at the sides of the pulse. When backprojected, these negative regions counteract the blur.

The fourth method is called **Fourier reconstruction**. In the spatial domain, CT reconstruction involves the relationship between a two-dimensional image and its set of one-dimensional views. By taking the two-dimensional Fourier transform of the image and the one-dimensional Fourier transform of each of its views, the problem can be examined in the frequency domain. As it turns out, the relationship between an image and its views is far simpler in the frequency domain than in the spatial domain. The frequency domain analysis